

Customer Information Management System - AWS Cost Analysis

Executive Summary

This comprehensive cost analysis provides detailed pricing estimates for the Customer Information Management System built on AWS DynamoDB. The system uses a serverless architecture with on-demand billing to optimize costs for variable usage patterns.

System Architecture Overview

The Customer Information Management System consists of: - **Frontend:** React.js web application (local development) - **Backend:** Node.js Express API server (local development) - **Database:** Amazon DynamoDB with on-demand billing - **Security:** Server-side encryption enabled - **Backup:** Point-in-time recovery (PITR) enabled

AWS Services and Pricing

Amazon DynamoDB - Primary Database Service

Core Pricing Components

Component	Unit Price	Description
Read Request Units	\$0.125 per million requests	Standard read operations
Write Request Units	\$0.625 per million requests	Create, update, delete operations
Storage (First 25 GB)	\$0.00 per GB-month	Free tier storage
Storage (Beyond 25 GB)	\$0.25 per GB-month	Additional storage
Point-in-Time Recovery	\$0.20 per GB-month	Backup storage

Global Secondary Index (GSI) Pricing

- **Read Request Units:** \$0.125 per million requests (same as table)
- **Write Request Units:** \$0.625 per million requests (same as table)
- **Storage:** \$0.25 per GB-month (no free tier for GSI)

Cost Scenarios Analysis

Scenario 1: Small Business (1,000 Customers)

Usage Assumptions: - 1,000 customer records - 5 GB total storage - 100,000 read operations/month - 20,000 write operations/month - 5 GB PITR backup storage

Monthly Cost Breakdown:

Storage: 5 GB (within free tier)	= \$0.00
Read Operations: $0.1M \times \$0.125/M$	= \$0.01
Write Operations: $0.02M \times \$0.625/M$	= \$0.01
PITR Backup: $5\text{ GB} \times \$0.20$	= \$1.00
GSI Storage: $\sim 0.5\text{ GB} \times \0.25	= \$0.13
GSI Operations: Minimal	= \$0.01

Total Monthly Cost:	= \$1.16
Annual Cost:	= \$13.92

Scenario 2: Medium Business (10,000 Customers)

Usage Assumptions: - 10,000 customer records - 50 GB total storage - 1,000,000 read operations/month - 200,000 write operations/month - 50 GB PITR backup storage

Monthly Cost Breakdown:

Storage: 25 GB free + $25\text{ GB} \times \$0.25$	= \$6.25
Read Operations: $1M \times \$0.125/M$	= \$0.13
Write Operations: $0.2M \times \$0.625/M$	= \$0.13
PITR Backup: $50\text{ GB} \times \$0.20$	= \$10.00
GSI Storage: $\sim 5\text{ GB} \times \0.25	= \$1.25
GSI Operations: $\sim 0.1M\text{ reads} \times \$0.125/M$	= \$0.01

Total Monthly Cost:	= \$17.77
Annual Cost:	= \$213.24

Scenario 3: Large Enterprise (100,000 Customers)

Usage Assumptions: - 100,000 customer records - 500 GB total storage - 10,000,000 read operations/month - 2,000,000 write operations/month - 500 GB PITR backup storage

Monthly Cost Breakdown:

Storage: 25 GB free + $475\text{ GB} \times \$0.25$	= \$118.75
Read Operations: $10M \times \$0.125/M$	= \$1.25
Write Operations: $2M \times \$0.625/M$	= \$1.25
PITR Backup: $500\text{ GB} \times \$0.20$	= \$100.00

GSI Storage: ~50 GB × \$0.25	= \$12.50
GSI Operations: ~1M reads × \$0.125/M	= \$0.13

Total Monthly Cost:	= \$233.88
Annual Cost:	= \$2,806.56

Cost Optimization Strategies

Immediate Optimizations

- 1. Monitor Usage Patterns**
 - Track actual read/write operations for 3 months
 - Evaluate if provisioned capacity would be more cost-effective
 - Potential savings: 60-80% for predictable workloads
- 2. Optimize Query Patterns**
 - Use partition keys efficiently to minimize hot partitions
 - Implement projection expressions to retrieve only necessary attributes
 - Use batch operations to reduce request units
- 3. Client-Side Caching**
 - Cache frequently accessed customer data
 - Reduce read request units by 30-50%
 - Implement TTL-based cache invalidation

Long-Term Optimizations

- 1. Reserved Capacity** (for predictable workloads)
 - Up to 76% savings on read/write capacity
 - Requires 1-year commitment
 - Best for stable, predictable usage patterns
- 2. DynamoDB Standard-IA** (Infrequent Access)
 - \$0.10 per GB-month storage (60% savings)
 - Higher read/write costs: \$0.155/\$0.78 per million requests
 - Suitable for archived or rarely accessed customer data
- 3. Auto Scaling** (for provisioned capacity)
 - Automatically adjust capacity based on demand
 - Prevent over-provisioning during low usage periods
 - Maintain performance during traffic spikes

Regional Cost Comparison

Region	Read RRU	Write RRU	Storage	PITR
US East (N. Virginia)	\$0.125/M	\$0.625/M	\$0.25/GB	\$0.20/GB
US West (Oregon)	\$0.125/M	\$0.625/M	\$0.25/GB	\$0.20/GB
EU (Ireland)	\$0.138/M	\$0.688/M	\$0.275/GB	\$0.22/GB
Asia Pacific (Tokyo)	\$0.148/M	\$0.738/M	\$0.295/GB	\$0.236/GB

Note: US East (N. Virginia) typically offers the lowest pricing

Cost Monitoring and Alerts

Recommended CloudWatch Alarms

1. **Monthly Cost Threshold**
 - Set billing alerts at 80% of expected monthly cost
 - Configure SNS notifications for cost overruns
2. **Request Unit Consumption**
 - Monitor read/write request unit consumption
 - Alert on unusual spikes that may indicate inefficient queries
3. **Storage Growth Rate**
 - Track storage growth trends
 - Plan for capacity and cost scaling

Cost Allocation Tags

Implement the following tags for cost tracking: - **Environment:** production, staging, development - **Application:** customer-management-system - **Owner:** team or department responsible - **CostCenter:** for chargeback purposes

Risk Assessment and Mitigation

Cost Risks

1. **Unexpected Traffic Spikes**
 - **Risk:** Sudden increase in read/write operations
 - **Mitigation:** Implement rate limiting and caching
2. **Data Growth**
 - **Risk:** Rapid storage growth beyond projections
 - **Mitigation:** Implement data lifecycle policies
3. **Inefficient Queries**
 - **Risk:** Hot partitions and scan operations
 - **Mitigation:** Regular query pattern reviews and optimization

Technical Risks

1. **Throttling**
 - **Risk:** Performance degradation during high load
 - **Mitigation:** Implement exponential backoff and retry logic
2. **Data Loss**
 - **Risk:** Accidental data deletion or corruption
 - **Mitigation:** PITR enabled, regular backup testing

Recommendations Summary

Phase 1: Initial Deployment (Months 1-3)

- Start with on-demand billing for flexibility
- Implement basic monitoring and alerting
- Optimize query patterns and implement caching
- **Expected Cost:** \$1-20/month depending on usage

Phase 2: Growth Phase (Months 4-12)

- Evaluate provisioned capacity based on usage patterns
- Consider reserved capacity for predictable workloads
- Implement advanced monitoring and cost optimization
- **Expected Cost:** \$15-200/month depending on growth

Phase 3: Scale Phase (Year 2+)

- Implement DynamoDB Standard-IA for archived data
- Consider multi-region deployment if needed
- Advanced cost optimization strategies
- **Expected Cost:** \$50-500+/month for enterprise scale

Conclusion

The Customer Information Management System on AWS DynamoDB provides a cost-effective, scalable solution with predictable pricing. Starting costs are minimal due to the free tier, and the on-demand billing model ensures you only pay for what you use. With proper optimization strategies, the system can scale efficiently while maintaining cost control.

Key Takeaways: - Start small with on-demand billing (~\$1-5/month for small businesses) - Scale predictably with clear cost projections - Significant optimization opportunities available as usage grows - Total cost of ownership remains competitive compared to traditional database solutions

This analysis is based on AWS pricing as of November 2025 and actual costs may vary based on usage patterns, regional pricing, and AWS pricing changes.